Estimation of the Conditional Beta of a Financial Asset

Master 104 - Research in Finance

Time Series Project 2025-2026

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1 Data

1.1 Data Sources and Time Period

We use daily adjusted closing prices for:

- Apple Inc. (AAPL): an individual equity listed on NASDAQ.
- S&P 500 Index (ˆGSPC): a broad market index representing the market portfolio.

The data are downloaded from Yahoo Finance via the `quantmod` R package. The sample spans from January 3, 2022 to the most recent available trading day, ensuring that the entire post-COVID recovery period is captured while avoiding contamination from the initial pandemic shock of 2020–2021.

Daily continuously compounded (log-) returns are computed as:

$$r_t = \ln P_t - \ln P_{t-1}, \tag{1}$$

where P_t denotes the adjusted closing price at date t.

1.2 Descriptive Statistics

Table 1 summarizes the key distributional characteristics of the daily log-returns.

Table 1: Descriptive Statistics of Daily Log-Returns

Statistic	AAPL	S&P 500
Frequency	Daily	Daily
Start date	2022-01-04	2022-01-04
End date	2026-03-09	2026-03-09
No. observations	1047	1047
Mean	0.000361	0.000333
Std. deviation	0.017838	0.011078
Skewness	0.3029	0.0497
Kurtosis	9.1292	9.5037

The kurtosis exceeds 3 for both series (raw kurtosis; the Gaussian benchmark is 3), indicating leptokurtic (fat-tailed) distributions. Both series exhibit slight positive skewness in this sample, though the dominant feature is the heavy tails characteristic of financial returns.

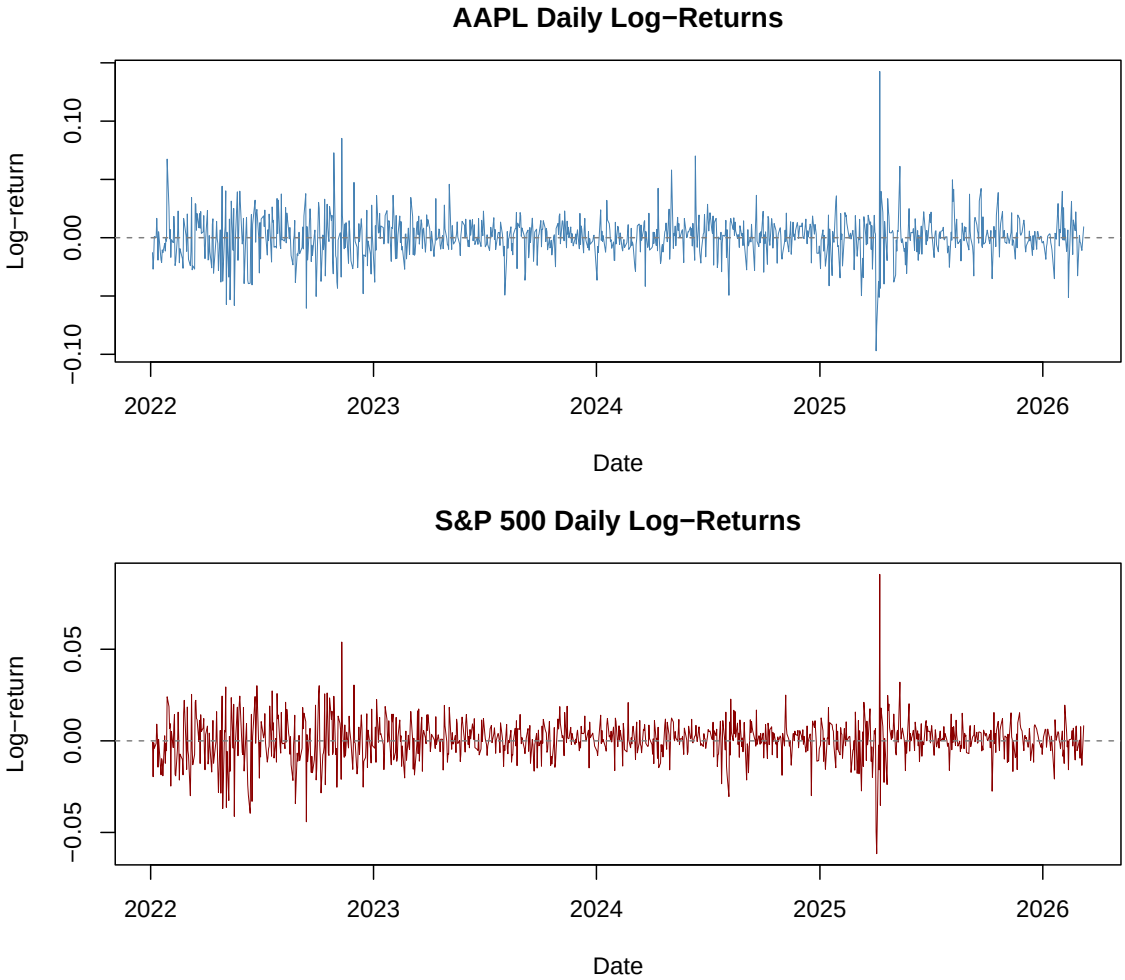

Figure 1: Daily log-returns of AAPL and S&P 500.

2 Methodology

2.1 Preliminary Analysis

2.1.1 Stationarity

We test for stationarity using the Augmented Dickey–Fuller (ADF) test. Under the null hypothesis of a unit root, the test regression is:

$$\Delta y_t = \alpha + \gamma\, y_{t-1} + \sum_{j=1}^{p} \delta_j\, \Delta y_{t-j} + \varepsilon_t. \tag{2}$$

Rejection of $H_0 : \gamma = 0$ implies stationarity.

2.1.2 Autocorrelation and ARCH Effects

We examine the autocorrelation function (ACF) of both the raw returns and the squared returns. The Ljung–Box test is used to test for serial correlation:

$$Q(m) = T(T+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{T-k} \sim \chi^2(m). \tag{3}$$

Engle's ARCH-LM test is applied to test for conditional heteroscedasticity in the residuals.

2.1.3 ARMA Modelling

If significant serial correlation is detected in the return series, we fit an ARMA(p,q) model for the conditional mean:

$$r_t = \mu + \sum_{i=1}^{p}\phi_i\, r_{t-i} + \sum_{j=1}^{q}\theta_j\, \varepsilon_{t-j} + \varepsilon_t. \tag{4}$$

Model selection is performed using the Bayesian Information Criterion (BIC).

2.2 Unconditional Beta

The unconditional (static) beta is estimated as:

$$\beta^i = \frac{\mathrm{Cov}(r_{i,t},\, r_{m,t})}{\mathrm{Var}(r_{m,t})}, \tag{5}$$

which corresponds to the OLS slope coefficient from regressing $r_{i,t}$ on $r_{m,t}$.

The unconditional annualized systematic risk is defined as:

$$SR^i = |\beta^i| \cdot \sigma_m \cdot \sqrt{A}, \tag{6}$$

where σ_m is the unconditional standard deviation of market returns and $A = 252$ is the annualization factor for daily data.

2.3 Univariate GARCH Model

For each return series r_t, we fit a GARCH(1,1) model with Student-t innovations:

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t\, z_t, \qquad z_t \sim t_\nu(0,1), \tag{7}$$

$$\sigma_t^2 = \omega + \alpha_1\, \varepsilon_{t-1}^2 + \beta_1\, \sigma_{t-1}^2, \tag{8}$$

with the stationarity constraint $\alpha_1 + \beta_1 < 1$.

2.4 DCC-GARCH Model

The Dynamic Conditional Correlation (DCC) model of proceeds in two stages.

Stage 1. Fit univariate GARCH models to each series and obtain standardized residuals:

$$\eta_t = D_t^{-1}\,\varepsilon_t, \tag{9}$$

where $D_t = \mathrm{diag}(\sigma_{1,t},\, \sigma_{2,t})$.

Stage 2. Model the conditional correlation via:

$$Q_t = (1-a-b)\,\bar{Q} + a\,\eta_{t-1}\,\eta_{t-1}' + b\,Q_{t-1}, \tag{10}$$

$$R_t = \mathrm{diag}(Q_t)^{-1/2}\,Q_t\,\mathrm{diag}(Q_t)^{-1/2}, \tag{11}$$

where $\bar{Q} = E[\eta_t\,\eta_t']$ is the unconditional correlation matrix of the standardized residuals, and $a, b \geq 0$ with $a + b < 1$.

The conditional covariance matrix is then:

$$H_t = D_t\,R_t\,D_t. \tag{12}$$

2.5 Conditional Beta and Systematic Risk

From the conditional covariance matrix H_t we extract:

- $\sigma_{im,t} = H_t[1,2]$: the conditional covariance between the asset and the market,
- $\sigma_{m,t}^2 = H_t[2,2]$: the conditional variance of the market.

The conditional beta is:

$$\boxed{\beta_t^i = \frac{\sigma_{im,t}}{\sigma_{m,t}^2},} \tag{13}$$

and the annualized conditional systematic risk is:

$$\boxed{SR_t^i = |\beta_t^i| \cdot \sigma_{m,t} \cdot \sqrt{A}.} \tag{14}$$

3 Empirical Results

3.1 Preliminary Results

3.1.1 Stationarity

The ADF tests strongly reject the null hypothesis of a unit root for both return series, confirming stationarity (Table 2).

Table 2: Augmented Dickey–Fuller Tests

Series	ADF Statistic	p-value	Conclusion
AAPL	-10.5640	< 0.01	Stationary
S&P 500	-9.8832	< 0.01	Stationary

3.1.2 Autocorrelation and ARCH Effects

The ACF of raw returns shows negligible autocorrelation, while the ACF of squared returns exhibits significant persistence, indicating volatility clustering (Figure 2).

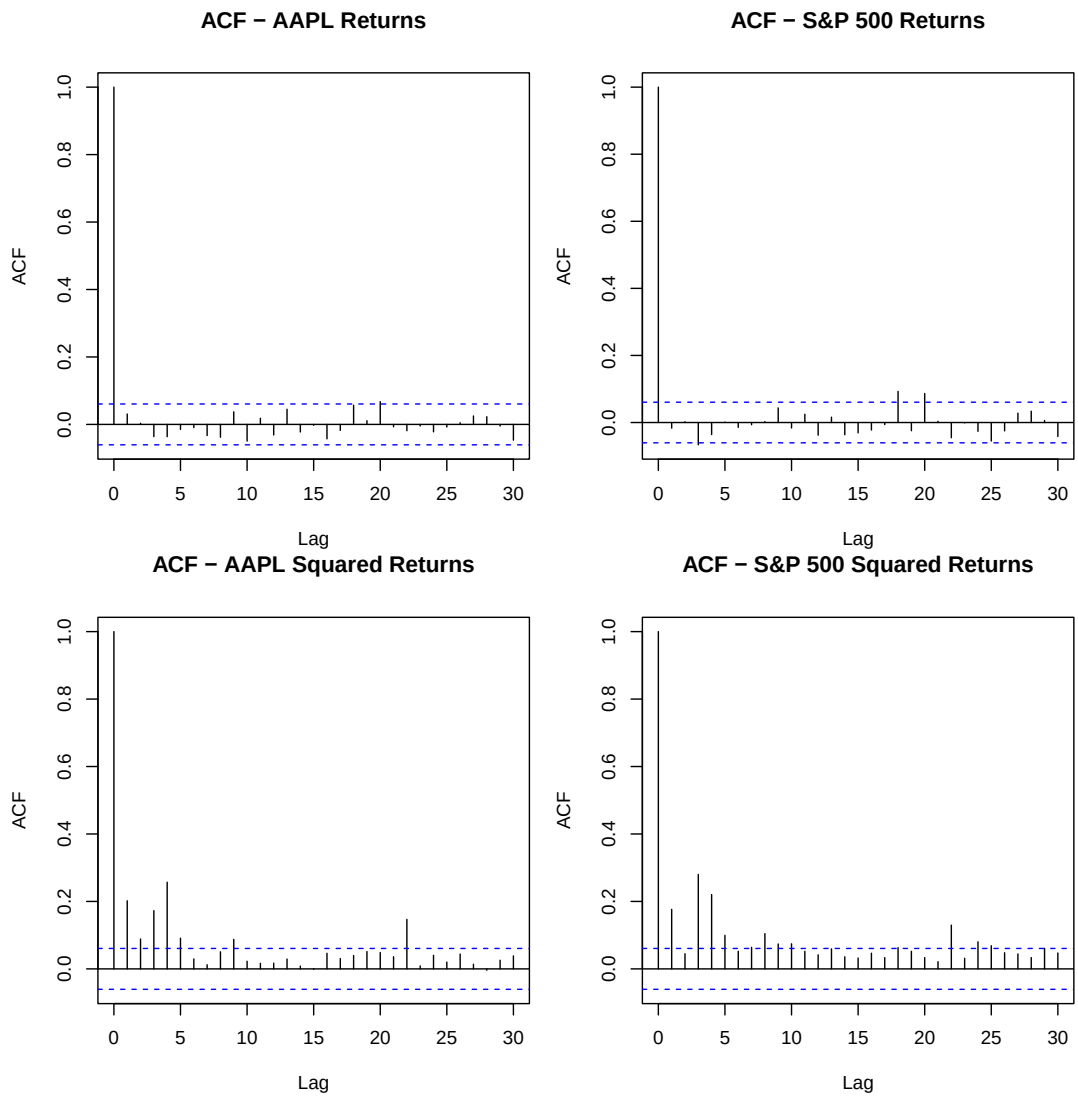

Figure 2: ACF of returns and squared returns.

Formal test results are presented in Table 3. The Ljung–Box test on raw returns yields non-significant p-values, indicating no serial correlation in the mean. On squared returns, the test strongly rejects, confirming volatility clustering. Engle's ARCH-LM test similarly rejects the absence of ARCH effects.

Table 3: Ljung–Box and ARCH-LM Tests (10 lags)

Test	Series	Statistic	p-value
Ljung–Box (returns)	AAPL	10.88	0.3666
	S&P 500	8.87	0.5441
Ljung–Box (squared)	AAPL	173.10	< 0.001
	S&P 500	209.29	< 0.001
ARCH-LM	AAPL	118.19	< 0.001
	S&P 500	139.90	< 0.001

3.1.3 Jarque–Bera Normality Test

The Jarque–Bera test strongly rejects the null of Gaussian returns for both series (Table 4), consistent with the observed leptokurtosis and motivating the choice of Student-t innovations in the GARCH specification.

Table 4: Jarque–Bera Normality Tests

Series	JB Statistic	p-value
AAPL	1654.85	< 0.001
S&P 500	1845.69	< 0.001

3.1.4 ARMA Model Selection

The BIC-based automatic ARMA selection yields ARMA(0,0) for both return series. This is consistent with the Ljung–Box results showing no significant serial correlation in the level of returns. Accordingly, the mean equation in the GARCH models reduces to a simple constant ($r_t = \mu + \varepsilon_t$).

3.2 Unconditional Estimates

The unconditional beta and systematic risk are reported in Table 5.

Table 5: Unconditional Estimates

Measure	Value
$\hat{\beta}^{\text{AAPL}}$	1.2261
Unconditional σ_m (ann.)	0.1759
SR^{AAPL} (ann.)	0.2156
R^2 (OLS)	0.5798

The $R^2 = 0.58$ indicates that approximately 58% of AAPL's daily return variation is explained by market movements.

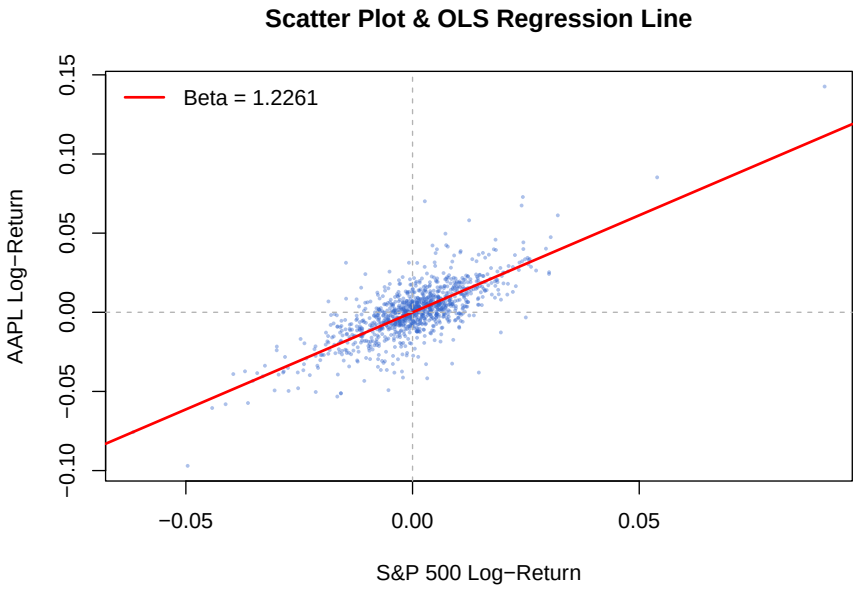

Figure 3: Scatter plot of AAPL vs. S&P 500 daily returns with OLS regression line.

3.3 Univariate GARCH Results

The GARCH(1,1) estimates for both series are summarized in Table 6.

Table 6: GARCH(1,1) Parameter Estimates

Parameter	AAPL	S&P 500
$\hat{\omega}$	6.447×10^{-6}	2.101×10^{-6}
$\hat{\alpha}_1$	0.0686	0.0996
$\hat{\beta}_1$	0.9150	0.8853
$\hat{\nu}$ (d.f.)	4.4258	6.9166
Persistence	0.9835	0.9849

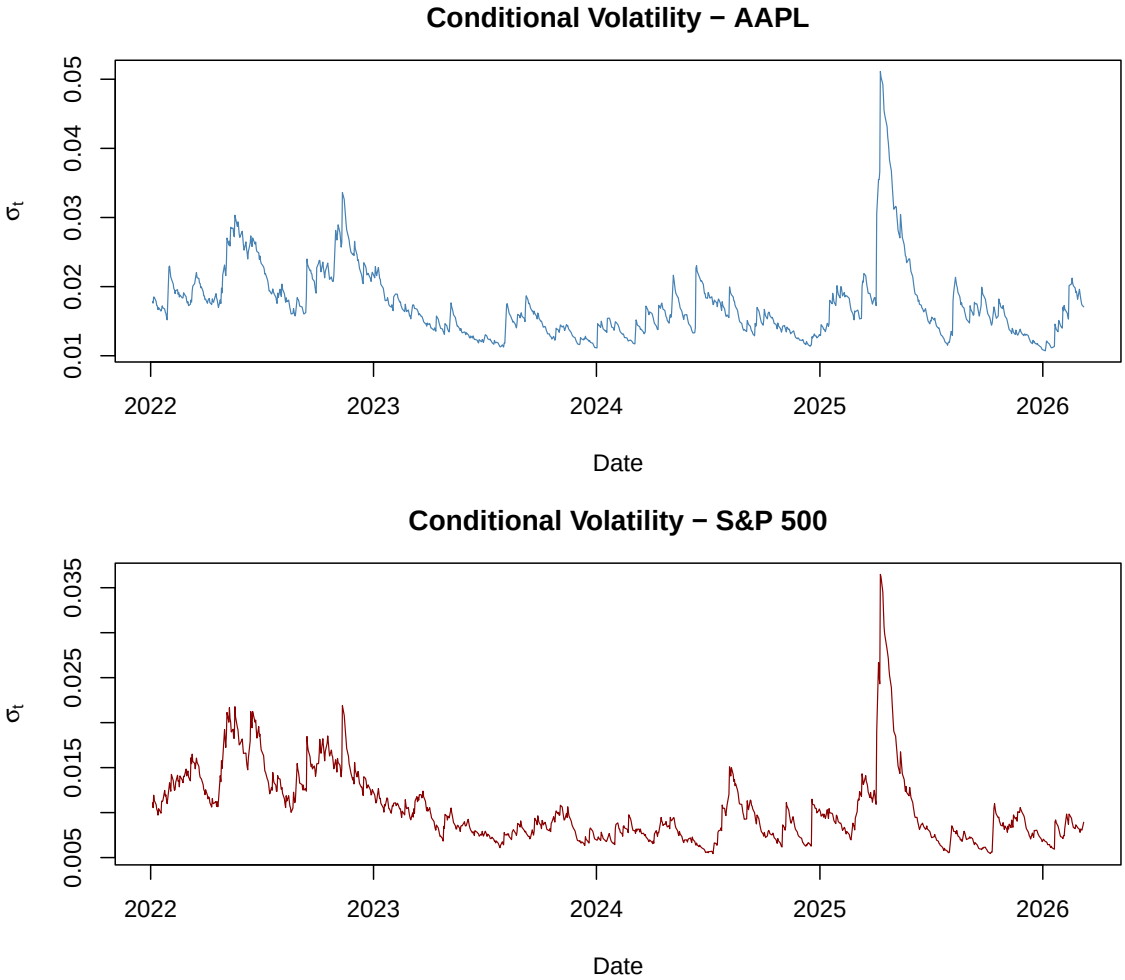

Figure 4: Conditional volatility from univariate GARCH(1,1) models.

3.4 DCC-GARCH Results

The DCC(1,1) parameters are reported in Table 7.

Table 7: DCC(1,1) Parameter Estimates

Parameter	Estimate
$\hat{a}$	0.0373
$\hat{b}$	0.9565
$\hat{a}+\hat{b}$	0.9938

The sum $\hat{a}+\hat{b}$ close to 1 indicates high persistence in the dynamic conditional correlations. The dominant role of $\hat{b}$ implies strong persistence in conditional correlation: today's correlation is primarily driven by yesterday's correlation, with only a modest correction toward the long-run mean. Conversely, the small $\hat{a}$ indicates that fresh return innovations update the correlation slowly, so the DCC process exhibits gradual rather than abrupt

adjustments. This pattern of slow mean-reversion has a direct bearing on the conditional beta series: beta episodes (either high or low) tend to last for extended periods rather than reverting quickly, which has practical implications for dynamic hedging strategies that rely on short-horizon beta estimates.

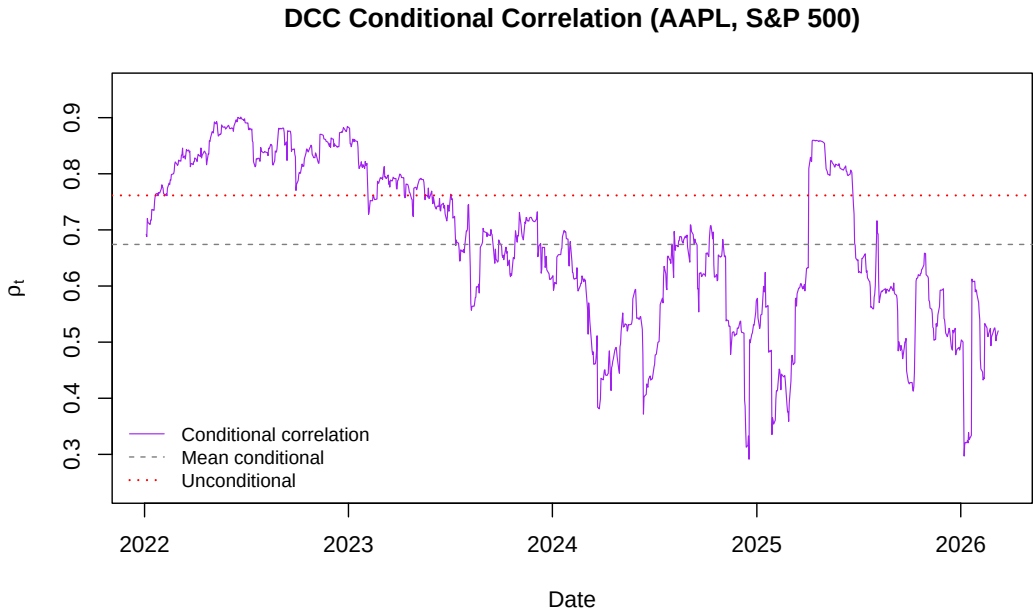

Figure 5: DCC conditional correlation between AAPL and S&P 500.

3.5 Conditional Beta

Figure 6 plots the conditional beta of AAPL over time.

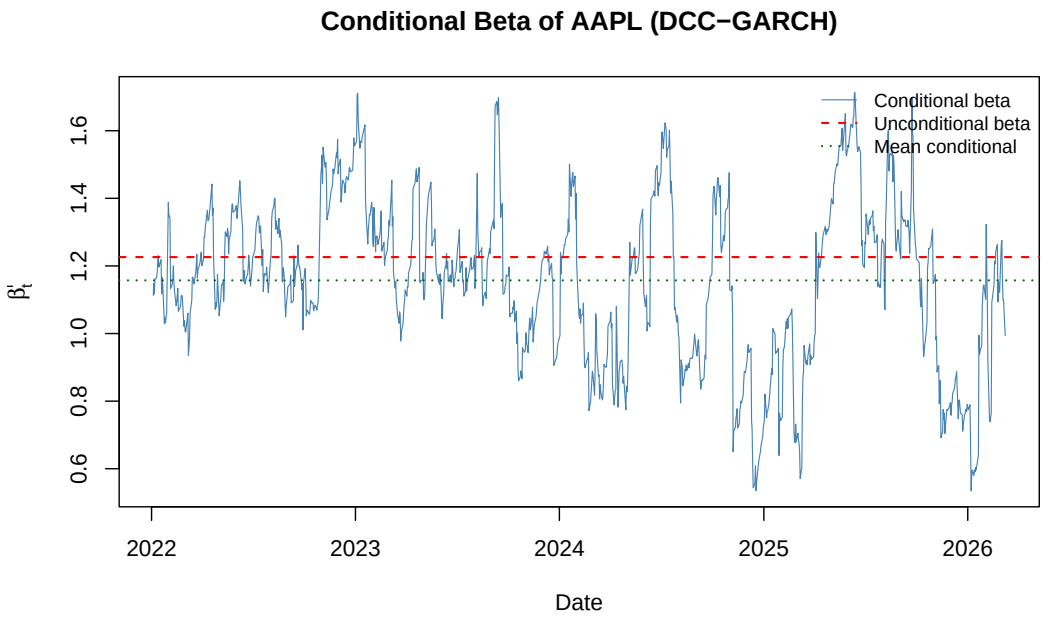

Figure 6: Conditional beta of AAPL estimated via DCC-GARCH.

The conditional beta fluctuates substantially around the unconditional OLS estimate, confirming that a constant-beta assumption is overly restrictive. Periods of market stress tend to coincide with elevated beta values, consistent with the "correlation breakdown" phenomenon.

3.6 Conditional Systematic Risk

Figure 7 displays the annualized conditional systematic risk.

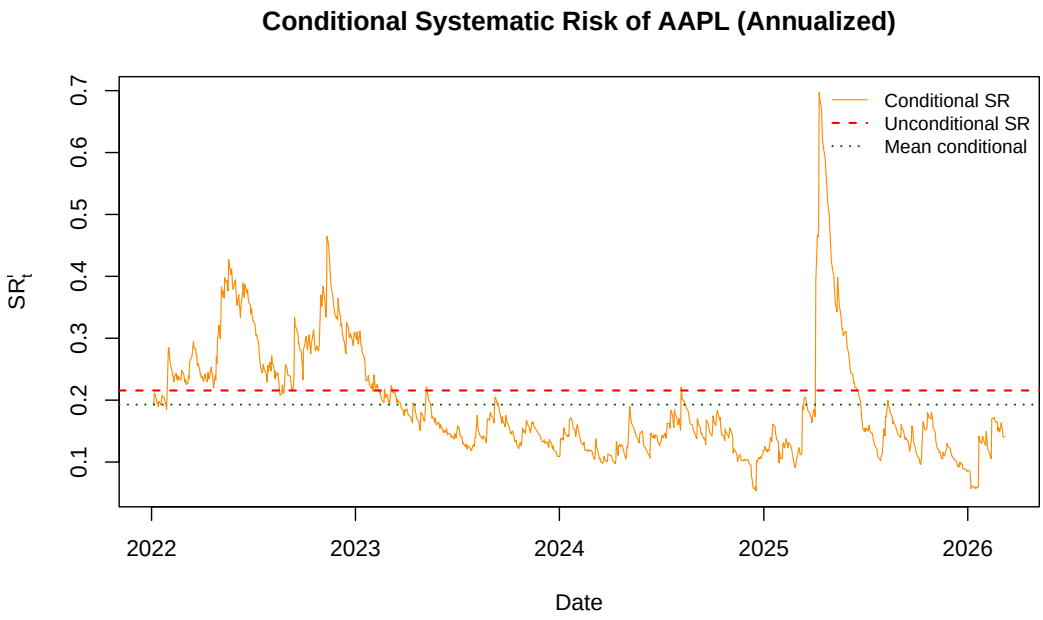

Figure 7: Annualized conditional systematic risk of AAPL.

3.7 Unconditional vs. Conditional: Summary

Table 8 contrasts the unconditional and conditional estimates.

Table 8: Comparison of Unconditional and Conditional Estimates

Measure	Unconditional	Conditional (mean $\pm$ std)
β	1.2261	1.1575 ± 0.2391
β range	—	$[0.5345,\ 1.7126]$
SR (annualized)	0.2156	0.1928 ± 0.0920

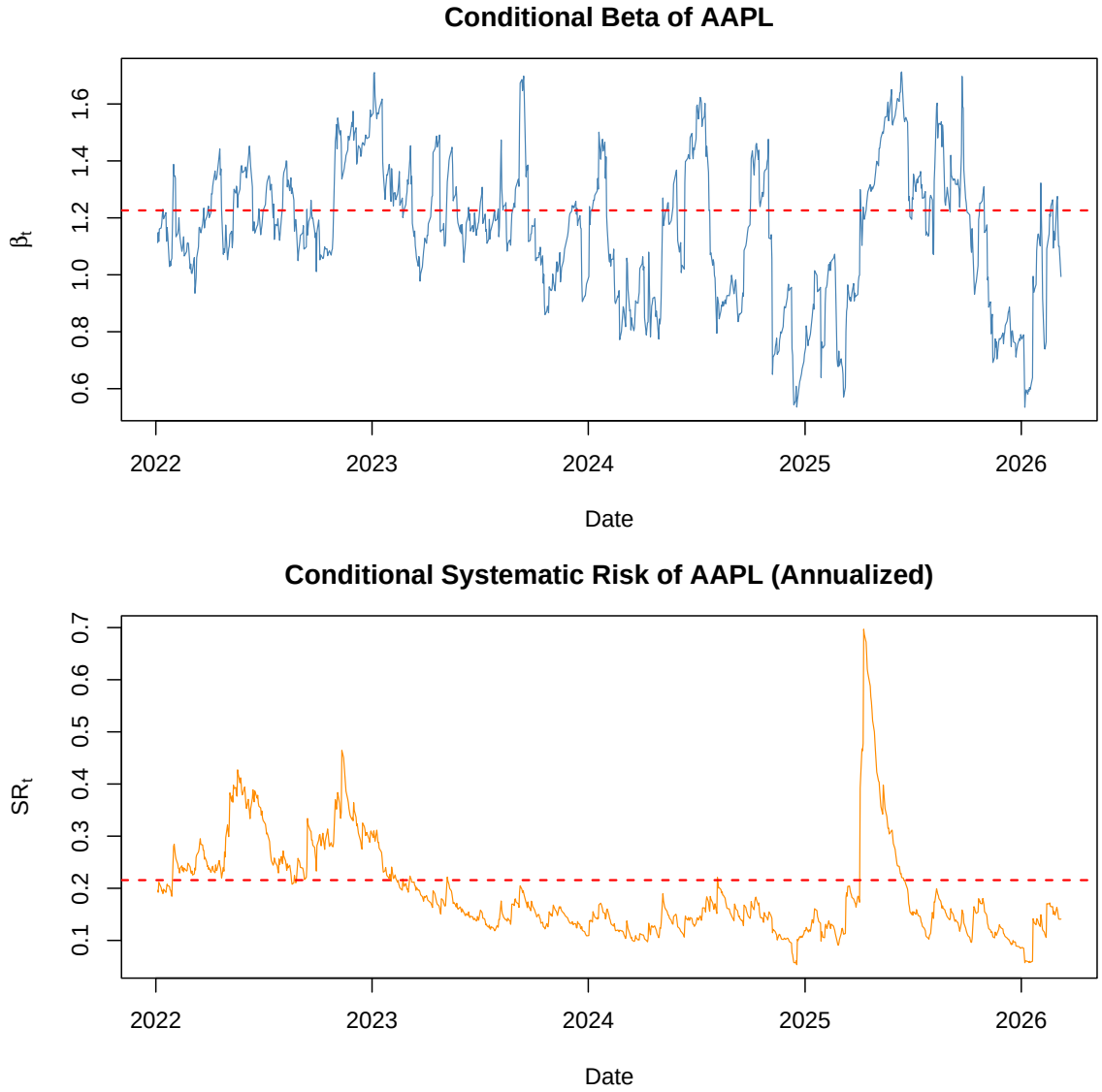

Figure 8: Conditional beta (top) and conditional systematic risk (bottom) of AAPL. The red dashed line indicates the unconditional estimate.

4 Conclusion

This study provides empirical evidence that the market beta of Apple Inc. is far from constant. Using a DCC-GARCH(1,1) framework applied to daily post-COVID data (January 2022 onward), we find:

1. The unconditional beta estimated via OLS provides a single point estimate that masks significant temporal variation.
2. The conditional beta, derived from the DCC-GARCH conditional covariance matrix, fluctuates considerably over time, with episodes of both elevated and depressed market sensitivity.
3. The conditional systematic risk similarly varies, driven by changes in both beta and

market volatility.

These findings have important implications for risk management and portfolio construction. A static beta can underestimate risk during periods of market turbulence and overestimate it during calm periods. The DCC-GARCH approach provides a parsimonious yet flexible alternative that captures the dynamic nature of systematic risk.

Future extensions could include asymmetric GARCH specifications (e.g., GJR-GARCH or EGARCH) to capture leverage effects.